A QUANTITATIVE PROOF OF STABLE PHASE RETRIEVAL

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Abstract. We give a fully explicit stability bound for phase retrieval in spans of independent real-valued random variables under a uniform two-sided L^1 bound. If $\xi_1, \ldots, \xi_n$ are independent, centered, unit-variance random variables satisfying $A \le \mathbb{E}[|\xi_i|] \le B$ for fixed $0 < A \le B < 1$, then for any orthogonal unit vectors $a, b \in \mathbb{R}^n$ the modulus gap $\sqrt{\mathbb{E}[(|X_a| - |X_b|)^2]}$ is at least $c_\perp(A, B) > 0$, where $c_\perp$ depends only on A and B.

This result is the quantitative counterpart of the compactness proof of stable phase retrieval [3]. Each step of the compactness argument—bounded probes, profile decomposition, head identification, tail rigidity—is replaced by an explicit finite-dimensional estimate with computable constants. The entire proof has been formalized in Lean 4 using the Mathlib library; Lean theorem names are recorded throughout.

1. Introduction

The phase retrieval problem asks whether a vector or function can be recovered from phaseless measurements. In the real setting this means recovery up to global sign. The finite-dimensional theory is by now extensive; see [4, 8] and the survey [10]. In infinite dimensions the situation is more delicate: phase retrieval can hold in infinite-dimensional Hilbert spaces [5], but uniform stability may fail for continuous frames [1], and natural measurement models such as Gabor frames may be severely ill-posed [2].

We adopt the function-space point of view from [9]. Given a closed subspace E of a real function space X, we say that E does *stable phase retrieval* with constant C if

$$\min_{\sigma \in \{\pm 1\}} \|f - \sigma g\|_X \le C \left\| |f| - |g| \right\|_X \qquad \text{for all } f, g \in E.$$

The map $f \mapsto |f|$ loses sign information, so the left-hand side optimizes over a global sign flip; the inequality says that this is the only obstruction to recovery. The main results of [9] show that many order-continuous Banach lattices admit no infinite-dimensional subspace with this property, while [7, 9] provide positive examples in L^2-spaces. The orthogonal reduction of [9] is the starting point of our argument: to verify stable phase retrieval on a subspace, it suffices to check it for orthogonal pairs in the same two-dimensional span.

In [6], Calderbank, Daubechies, Freeman, and Freeman proved stable phase retrieval for a model involving shifts and proposed the following conjecture: if $(\xi_i)_{i \ge 1}$ is an orthonormal sequence of independent mean-zero random variables in L^2, then $\overline{\mathrm{span}}(\xi_i)$ does stable phase retrieval if and only if there exist $0 < A \le B < 1$ with

$$A \le \|\xi_i\|_{L^1} \le B$$

for all but at most one i. The lower bound prevents ξ_i from concentrating near zero; the upper bound prevents it from being Rademacher-valued (in which case $|\xi_i|$ is constant and carries no phase

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information). Together they ensure that each coordinate contributes usable information about the signs of the coefficients.

Our main result proves the sufficiency direction of this conjecture with a fully explicit stability constant.

1.1. **The main theorem.** Let $\xi_1, \ldots, \xi_n$ be independent real-valued random variables on a probability space (Ω, μ), each satisfying

$$\mathbb{E}[\xi_i] = 0, \qquad \mathbb{E}[\xi_i^2] = 1, \qquad A \le \mathbb{E}[|\xi_i|] \le B$$

for fixed $0 < A \le B < 1$. Given unit vectors $a, b \in \mathbb{R}^n$ with $\langle a, b\rangle = 0$, form the random linear combinations $X_a = \sum_i a_i\xi_i$ and $X_b = \sum_i b_i\xi_i$. The *modulus gap*

$$\delta(a, b) = \sqrt{\mathbb{E}\big[(|X_a| - |X_b|)^2\big]}$$

measures how distinguishable X_a and X_b are from their absolute values alone. After the orthogonal reduction, stable phase retrieval with constant C is equivalent to $\delta(a, b) \ge \sqrt{2}/C$ for all orthogonal unit pairs.

Theorem 1.1 (Quantitative orthogonal lower bound)**.** `Showcase.quantitative_orthogonal_lower_bound`

There exists a constant $c_\perp(A, B) > 0$, which can be chosen to be

$$c_\perp(A, B) = \tfrac{1}{2}\min\big(c_{\text{cross}}(A),\ c_{\text{line}}(A, B)\big),$$

such that $c_\perp(A, B) \le \delta(a, b)$ for every pair of orthogonal unit vectors $a, b \in \mathbb{R}^n$. The constant depends only on A and B, not on the dimension n or the laws of ξ_i. The quantities c_{cross} and c_{line} are defined in Section 2.3.

Remark 1.2. The companion paper [3] proves the same conclusion (without explicit constants) by a compactness argument that also allows one exceptional variable with no L^1 bound. The present proof makes each step of that argument quantitative.

1.2. **Proof strategy.** Rather than lower-bounding δ directly, we work with the auxiliary quantity

$$\varepsilon(a, b) = \mathbb{E}\big[|X_a^2 - X_b^2|\big],$$

which is more amenable to algebraic manipulation. A Cauchy–Schwarz argument (Theorem A.6) gives $\varepsilon \le 2\delta$, so any lower bound on ε transfers to δ at the cost of a factor of two.

The proof splits the coordinates into a *head* (the few coordinates where a or b has a large coefficient) and a *tail* (the remaining coordinates, each individually small). A sign dichotomy on the tail (Lemma 3.1) then determines which of two branches applies:

(i) **Concentrated branch** (Section 5). For some sign $\tau = \pm 1$, the tail of $a - \tau b$ is small, so a and τb nearly agree outside the head. Bounded probe functions extract entries of the head difference matrix from the observable $X_a^2 - X_b^2$, each entry being controlled by $K(A, B) \cdot \varepsilon$. The orthogonality $\langle a, b\rangle = 0$ forces this matrix to have large Frobenius norm, giving $\varepsilon \ge c_{\text{line}}(A, B)$.

(ii) **Diffuse branch** (Section 6). Both $a + b$ and $a - b$ have substantial tail energy, spread across many small coefficients. A greedy block partition, symmetrization via the product space, Sperner's antichain bound, and a Chebyshev counting argument together show that X_a and X_b are both unlikely to be near zero, giving $\varepsilon \ge c_{\text{cross}}(A)$.

In both cases $\varepsilon \ge \min(c_{\text{cross}}, c_{\text{line}})$, and Theorem 1.1 follows.

2. Setup: the head–tail decomposition

Throughout, $a, b \in \mathbb{R}^n$ are unit vectors with $\langle a, b\rangle = 0$, and $\xi_1, \dots, \xi_n$ are independent random variables satisfying the moment conditions of Section 1. We write $X_c = \sum_i c_i \xi_i$ for any coefficient vector $c \in \mathbb{R}^n$.

The first step is to separate the coordinates where a and b have large coefficients from those where they are small. This is the quantitative counterpart of the profile decomposition in [3]: the large coordinates form a finite set on which we can extract detailed information via probes, while the small coordinates contribute a "diffuse" component that is controlled by anticoncentration.

Definition 2.1 (Head set, restriction, and tail)**.** `Core.FiniteModel.headSetFin`

The *head set* is $H = H(A, a, b) = \{i : \max(|a_i|, |b_i|) > \lambda(A)\}$. For $S \subseteq \{1, \dots, n\}$ and $c \in \mathbb{R}^n$, write $\mathrm{head}_S\, c$ for the vector that agrees with c on S and vanishes elsewhere (`restrictCoeff`), and $\mathrm{tail}_S\, c = c - \mathrm{head}_S\, c$ for its complement (`tailCoeff`).

Since each $i \in H$ contributes at least $\lambda(A)^2$ to $a_i^2 + b_i^2 \le 2$, the head set is finite with $|H| \le 2/\lambda(A)^2 \le d^*(A)$ (`headSetFin_card_le_dStar`). Conversely, every tail coordinate satisfies $\max(|(\mathrm{tail}_H\, a)_i|, |(\mathrm{tail}_H\, b)_i|) \le \lambda(A)$ by construction (`headSetFin_tail_max_le_lambda`).

2.1. The head difference matrix.

The head difference matrix captures the discrepancy between a and b on the large coordinates. It is a rank-two symmetric matrix whose Frobenius norm is forced to be large by the orthogonality constraint whenever the tails are small.

Definition 2.2 (Difference matrix)**.** `Core.FiniteModel.finDH`

The head difference matrix $D_H(a, b)$ is the symmetric matrix with entries

$$D_H(a, b)_{ij} = (\mathrm{head}_H\, a)_i (\mathrm{head}_H\, a)_j - (\mathrm{head}_H\, b)_i (\mathrm{head}_H\, b)_j$$

and Frobenius norm $\|D_H\|_F = \sqrt{\sum_{i,j} D_H(a, b)_{ij}^2}$.

2.2. The epsilon–modulus bridge.

The two key quantities measuring phase-retrieval distinguishability are

$$\varepsilon(a, b) = \mathbb{E}\big[|X_a^2 - X_b^2|\big], \qquad \delta(a, b) = \sqrt{\mathbb{E}\big[(|X_a| - |X_b|)^2\big]}.$$

Since $|X_a^2 - X_b^2| = |X_a + X_b| \cdot |X_a - X_b|$, Cauchy–Schwarz and the variance identity $\mathbb{E}[X_c^2] = \sum c_i^2$ (`Imported.L2Theory.randL2Sq_finLinComb_eq`) yield $\varepsilon \le 2\delta$ (Theorem A.6). The rest of the proof is devoted to showing ε is bounded below.

2.3. Table of constants.

All constants depend only on A and B and are explicitly computable. The probe bound, gluing threshold, and head/tail cutoff are

$$K = \max\big(\tfrac{4}{A^2}, \tfrac{2}{1-B}\big), \qquad \theta = \tfrac{A^2}{8}, \qquad \lambda = \tfrac{A^3}{2^{20}}.$$

The head set has at most $d^* = 2 + 2^{41}/A^6$ elements, and the diffuse threshold is $s_0 = A^4/2^{20}$. The two branch constants and the stability constant are

$$c_{\mathrm{cross}} = \tfrac{A^8}{2^{57}}, \qquad c_{\mathrm{line}} = \tfrac{A^2}{2^8\, d^* K}, \qquad c_\perp = \tfrac{1}{2}\min(c_{\mathrm{cross}}, c_{\mathrm{line}}).$$

The block partition in Section 6 also uses

$$M = \lceil 2^{18}/A^2 \rceil, \quad \eta(s) = \tfrac{s}{4M}, \quad \lambda_0(s) = \tfrac{A\sqrt{s}}{2048}, \quad r_0(s) = \tfrac{A^2\sqrt{s}}{65536}.$$

3. The tail sign dichotomy

The vectors $a+b$ and $a-b$ partition the energy of the pair (a,b) into two complementary directions. On the tail H^c, at least one of these directions must carry significant energy—but it may be that both do, or that one is negligible. This dichotomy determines which branch of the proof applies.

Lemma 3.1 (Tail sign dichotomy). `Main.UniformAB.tail_sign_dichotomy`

Let $a,b \in \mathbb{R}^n$ be unit vectors with $\langle a,b\rangle = 0$ and set $H = H(A,a,b)$. Exactly one of the following holds:

(i) ***Diffuse:*** $\|\mathrm{tail}_H(a-b)\|_2^2 \geq 16\, s_0(A)$ *and* $\|\mathrm{tail}_H(a+b)\|_2^2 \geq 16\, s_0(A)$.
(ii) ***Concentrated:*** *there exists* $\tau \in \{+1,-1\}$ *with* $\|\mathrm{tail}_H(a-\tau b)\|_2 < \theta(A)/32$.

Proof. If $\|\mathrm{tail}_H(a-b)\|_2^2 < 16\, s_0$, take $\tau = 1$; if instead $\|\mathrm{tail}_H(a+b)\|_2^2 < 16\, s_0$, take $\tau = -1$. Since $16\, s_0 = A^4/2^{16} = (\theta/32)^2$, the squared bound yields the norm bound in (ii). If neither holds, both squared norms are at least $16\, s_0$ and we are in (i). □

Remark 3.2. In the concentrated case, a and τb nearly agree on the tail, so their difference is concentrated on the finite head set H. In the diffuse case, both $a \pm b$ have energy spread across many small tail coefficients, precisely the regime where anticoncentration is effective.

4. Auxiliary estimates

Before entering the two branches of the proof, we collect the three main tools: bounded probe functions for reading off coefficients from the quadratic observable $X_a^2 - X_b^2$; an L^1 lower bound for independent sums; and a gluing estimate showing that if a and τb nearly agree *everywhere* (head and tail), then ε is already large.

4.1. Probe functions.

Given a centered random variable ξ with $\mathbb{E}[|\xi|] > 0$, one can always find a bounded function of ξ that acts as a dual to ξ itself. If additionally ξ is not Rademacher-valued (i.e., $\mathbb{E}[|\xi|] < 1$), the same construction works for $\xi^2 - 1$. These are the probes that let us read off individual entries of D_H from ε.

Lemma 4.1 (Linear probe). `Imported.ElementaryBounds.exists_linear_probe`

Let ξ be a real random variable with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[|\xi|] \geq A > 0$. Then there exists a measurable $g : \mathbb{R} \to \mathbb{R}$ with $|g| \leq 2/A$, $\mathbb{E}[g(\xi)] = 0$, and $\mathbb{E}[\xi\, g(\xi)] = 1$.

Proof. Let $D = \mathbb{E}[|\xi|]$ and $c = \mathbb{E}[\mathrm{sign}(\xi)]$, and set $g(x) = (\mathrm{sign}(x) - c)/D$. Then $\mathbb{E}[g(\xi)] = 0$ and $\mathbb{E}[\xi\, g(\xi)] = \mathbb{E}[|\xi|]/D = 1$. The bound $|g| \leq 2/D$ follows from $|\mathrm{sign}(x) - c| \leq 2$. □

Lemma 4.2 (Quadratic probe). `Imported.ElementaryBounds.exists_quadratic_probe`

Let ξ be a real random variable with $\mathbb{E}[\xi] = 0$, $\mathbb{E}[\xi^2] = 1$, and $\mathbb{E}[|\xi|] \leq B < 1$. Then there exists a measurable $g : \mathbb{R} \to \mathbb{R}$ with $|g| \leq 2/(1-B)$, $\mathbb{E}[g(\xi)] = 0$, and $\mathbb{E}[(\xi^2 - 1)\, g(\xi)] = 1$.

Proof. Apply the linear-probe construction to $\eta = \xi^2 - 1$, whose L^1 norm satisfies $\mathbb{E}[|\xi^2 - 1|] \geq 1 - \mathbb{E}[|\xi|] \geq 1 - B > 0$. □

Remark 4.3. The linear probe recovers c_i from $X_c = \sum_k c_k \xi_k$ via the identity $\mathbb{E}[X_c \cdot g(\xi_i)] = c_i$: the centering $\mathbb{E}[g(\xi_i)] = 0$ kills the contribution of every ξ_k with $k \neq i$, and the normalization $\mathbb{E}[\xi_i g(\xi_i)] = 1$ picks out c_i. Likewise, the quadratic probe gives $\mathbb{E}[X_c^2 \cdot g(\xi_i)] = c_i^2$.

4.2. **The L^1 lower bound.** Independent sums of variables with positive L^1 norms cannot concentrate near zero. This is the engine behind both the gluing estimate and the diffuse branch: it converts ℓ^2 energy of the coefficients into L^1 mass of the random sum.

Theorem 4.4 (L^1 lower bound for independent sums)**.** `Imported.ElementaryBounds.finite_indep_sum_l1_lower`

Let $\zeta_1, \dots, \zeta_n$ be independent, mean-zero, unit-variance random variables with $\mathbb{E}[|\zeta_i|] \geq A > 0$ for all i. Then for every $(c_1, \dots, c_n) \in \mathbb{R}^n$,

$$\mathbb{E}\Big[\Big|\sum_{i=1}^{n} c_i \zeta_i\Big|\Big] \geq \frac{A}{4}\sqrt{\sum_{i=1}^{n} c_i^2}\,.$$

4.3. **The four-term gluing estimate.** Orthogonality forces $\|a+\tau b\|_2 = \|a-\tau b\|_2 = \sqrt{2}$. If for some τ both the head of $a-\tau b$ and the tail of $a+\tau b$ are small, then the complementary pieces—the tail of $a-\tau b$ and the head of $a+\tau b$—must be close to $\sqrt{2}$. The product $X_{a+\tau b} \cdot X_{a-\tau b} = X_a^2 - X_b^2$ then has a large main term and small errors, yielding a lower bound on ε.

Lemma 4.5 (Four-term bound)**.** `Tail.SignMismatch.fin_sign_mismatch_measure_bound`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$, let $H \subseteq \{1, \dots, n\}$, and let $\tau \in \{+1, -1\}$. Write $c^+ = a + \tau b$ and $c^- = a - \tau b$. Then

$$\begin{aligned}\varepsilon(a,b) \;\geq\; & \left(\tfrac{A}{4}\right)^2 \|\mathrm{head}_H\, c^+\|_2\, \|\mathrm{tail}_H\, c^-\|_2 \\ & - \|\mathrm{head}_H\, c^+\|_2\, \|\mathrm{head}_H\, c^-\|_2 - \|\mathrm{tail}_H\, c^+\|_2\, \|\mathrm{tail}_H\, c^-\|_2 \\ & - \|\mathrm{tail}_H\, c^+\|_2\, \|\mathrm{head}_H\, c^-\|_2.\end{aligned}$$

Proof. The identity $X_a^2 - X_b^2 = X_{c^+} \cdot X_{c^-}$ holds since $\tau^2 = 1$. Set $P = X_{\mathrm{head}_H\, c^+}$, $X' = X_{\mathrm{tail}_H\, c^+}$, $W = X_{\mathrm{head}_H\, c^-}$, $V = X_{\mathrm{tail}_H\, c^-}$. Expanding $(P + X')(W + V)$ and rearranging:

$$|PV| \leq |X_{c^+} X_{c^-}| + |PW| + |X'V| + |X'W|.$$

Taking expectations, the left-hand side factors as $\mathbb{E}[|P|] \cdot \mathbb{E}[|V|]$ by independence of disjoint supports (`Imported.L2Theory.indepFun_finLinComb_disjoint`). Each factor is at least $(A/4)$ times the ℓ^2 norm of the corresponding coefficient vector, by Theorem 4.4. The three error terms are each bounded by the product of the relevant L^2 norms via Cauchy–Schwarz. □

Proposition 4.6 (Gluing)**.** `Tail.SignMismatch.fin_sign_mismatch_gluing`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$, let $H \subseteq \{1, \dots, n\}$, and let $\tau \in \{+1, -1\}$. If

$$\|\mathrm{head}_H(a - \tau b)\|_2 \leq \theta(A)/16 \quad \textit{and} \quad \|\mathrm{tail}_H(a + \tau b)\|_2 \leq \theta(A)/32,$$

then $\varepsilon(a, b) \geq \theta(A)/2$.

Proof. Write $c^+ = a + \tau b$ and $c^- = a - \tau b$. Set $\alpha = \|\mathrm{head}_H\, c^-\|_2$ and $\beta = \|\mathrm{tail}_H\, c^+\|_2$; by hypothesis $\alpha \leq \theta/16$ and $\beta \leq \theta/32$. From $\|c^\pm\|_2^2 = 2$ we get

$$\|\mathrm{head}_H\, c^+\|_2 = \sqrt{2 - \beta^2} \geq 1, \qquad \|\mathrm{tail}_H\, c^-\|_2 = \sqrt{2 - \alpha^2} \geq 1.$$

The main term in the four-term bound (Lemma 4.5) is therefore at least $(A/4)^2 \cdot 1 \cdot 1 = \theta/2$, while the three error terms sum to at most $\alpha\sqrt{2} + \beta\sqrt{2} + \alpha\beta < \theta/4$. The conclusion $\varepsilon \geq \theta/2$ follows. □

5. The concentrated branch

In the concentrated case of the dichotomy (Lemma 3.1), there exists $\tau = \pm 1$ such that a and τb nearly agree on the tail. Their difference is therefore concentrated on the head set H, and the head difference matrix D_H must be large. The strategy is to show that every entry of D_H is controlled by $K \cdot \varepsilon$, while the Frobenius norm of D_H is forced to be at least 1 by orthogonality. Together these give a lower bound on ε. If ε is too small, a bootstrapping argument via the gluing estimate (Proposition 4.6) produces a contradiction.

5.1. **Entry bounds via probes.**

Proposition 5.1 (Head entry bound)**.** `Head.HeadMatrix.head_entry_bound_fin`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$ and let $H = H(A, a, b)$. For all $i, j \in H$,

$$|D_H(a, b)_{ij}| \le K(A, B) \cdot \varepsilon(a, b).$$

Proof. Diagonal entries. Let g be the quadratic probe for ξ_i (Lemma 4.2). Writing $X_c = c_i\xi_i + R$ with $R = \sum_{k\neq i} c_k\xi_k$ independent of ξ_i, the computation in Remark 4.3 gives $\mathbb{E}[X_c^2 \cdot g(\xi_i)] = c_i^2$. Applying this to both a and b:

$$|a_i^2 - b_i^2| = \left|\mathbb{E}\left[(X_a^2 - X_b^2)g(\xi_i)\right]\right| \le \frac{2}{1-B} \cdot \varepsilon \le K \cdot \varepsilon.$$

Off-diagonal entries. Let g_i, g_j be linear probes for ξ_i, ξ_j (Lemma 4.1). The same independence argument gives $\mathbb{E}[X_c^2 \cdot g_i(\xi_i)g_j(\xi_j)] = 2c_ic_j$, whence

$$|a_ia_j - b_ib_j| \le \tfrac{1}{2} \cdot (2/A)^2 \cdot \varepsilon = \frac{2}{A^2}\,\varepsilon \le K \cdot \varepsilon. \qquad \square$$

5.2. **Frobenius control.**

Lemma 5.2 (Frobenius lower bound)**.** `Head.HeadMatrix.head_dominant_lower_bound_fin`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$ and let $H \subseteq \{1, \dots, n\}$. If $\|\mathrm{tail}_H\, a\|_2^2 + \|\mathrm{tail}_H\, b\|_2^2 \le 1/100$, then $\|D_H(a, b)\|_F \ge 1$.

Proof. Write $\rho_a = \|\mathrm{tail}_H\, a\|_2^2$ and $\rho_b = \|\mathrm{tail}_H\, b\|_2^2$, so that $\|\mathrm{head}_H\, a\|_2^2 = 1 - \rho_a$ and $\|\mathrm{head}_H\, b\|_2^2 = 1 - \rho_b$. The global orthogonality $\langle a, b\rangle = 0$ splits as $\langle \mathrm{head}_H\, a, \mathrm{head}_H\, b\rangle = -\langle \mathrm{tail}_H\, a, \mathrm{tail}_H\, b\rangle$, and Cauchy–Schwarz gives $|\langle \mathrm{head}_H\, a, \mathrm{head}_H\, b\rangle| \le \sqrt{\rho_a\rho_b}$. Writing $z = \langle \mathrm{head}_H\, a, \mathrm{head}_H\, b\rangle$, the Frobenius norm satisfies

$$\|D_H\|_F^2 = (1-\rho_a)^2 + (1-\rho_b)^2 - 2z^2.$$

Since $\rho_a + \rho_b \le 1/100$ and $z^2 \le \rho_a\rho_b \le (\rho_a + \rho_b)^2/4$, one checks $\|D_H\|_F^2 \ge 1$. $\square$

Combining the entry bound with the Frobenius lower bound: if $\|\mathrm{tail}_H\, a\|_2^2 + \|\mathrm{tail}_H\, b\|_2^2 \le 1/100$, then $1 \le \|D_H\|_F^2 \le |H|^2(K\varepsilon)^2$, giving $\varepsilon \ge 1/(|H| \cdot K)$ (`Head.HeadMatrix.threshold_head_lower_bound_fin`).

5.3. **The one-line branch.** We now close the concentrated case. The key algebraic ingredient is:

Lemma 5.3 (Rank-one algebra)**.** `Tail.OneLine.rank_one_sign_selection`

For $x, y \in \mathbb{R}^m$, let $D(x, y)_{ij} = x_ix_j - y_iy_j$. Then

$$\|x - y\|_2^2 \cdot \|x + y\|_2^2 \le 2\,\|D(x, y)\|_F^2.$$

Proof. Writing $s = \|x\|_2^2$, $t = \|y\|_2^2$, $c = \langle x, y\rangle$, the left-hand side is $(s + t)^2 - 4c^2$ and the right is $2(s^2 + t^2 - 2c^2)$. The difference is $(s - t)^2 \ge 0$. $\square$

Lemma 5.4 (Head sign selection). `Tail.OneLine.head_sign_selection`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$, let $H \subseteq \{1, \dots, n\}$, and let $\tau \in \{+1, -1\}$. If $\|\mathrm{tail}_H(a - \tau b)\|_2 \le \theta(A)/32$ and every entry of D_H satisfies $|D_H(a, b)_{ij}| \le K(A, B) \cdot \varepsilon(a, b)$, then

$$\|\mathrm{head}_H(a + \tau b)\|_2 \le 2\,|H| \cdot K(A, B) \cdot \varepsilon(a, b).$$

Proof. Apply Lemma 5.3 to $x = \mathrm{head}_H\, a$, $y = \tau\ \mathrm{head}_H\, b$:

$$\|\mathrm{head}_H(a - \tau b)\|_2^2 \cdot \|\mathrm{head}_H(a + \tau b)\|_2^2 \le 2\,|H|^2 (K\varepsilon)^2.$$

Pythagoras gives $\|\mathrm{head}_H(a-\tau b)\|_2^2 = 2 - \|\mathrm{tail}_H(a-\tau b)\|_2^2 \ge 1$. Dividing through and taking square roots: $\|\mathrm{head}_H(a + \tau b)\|_2 \le \sqrt{2}\,|H| \cdot K\varepsilon \le 2\,|H| \cdot K\varepsilon$. □

Proposition 5.5 (One-line branch). `Tail.OneLine.one_line_branch`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$, let $H \subseteq \{1, \dots, n\}$ with $|H| \le d^*(A)$, and let $\tau \in \{+1, -1\}$. If $\|\mathrm{tail}_H(a - \tau b)\|_2 \le \theta(A)/32$, then $\varepsilon(a, b) \ge c_{\mathrm{line}}(A, B)$.

Proof. Suppose $\varepsilon < c_{\mathrm{line}} = A^2/(2^8 d^* K)$. By Proposition 5.1, every entry of D_H satisfies $|D_{ij}| \le K\varepsilon$, so Lemma 5.4 gives

$$\|\mathrm{head}_H(a + \tau b)\|_2 < 2d^* K \cdot \frac{A^2}{2^8 d^* K} = \frac{A^2}{128} = \frac{\theta}{16}.$$

Now swap the sign: the head of $a + \tau b$ has norm less than $\theta/16$, and the tail of $a - \tau b$ has norm at most $\theta/32$. The gluing estimate (Proposition 4.6) applied with sign $-\tau$ gives $\varepsilon \ge \theta/2 > c_{\mathrm{line}}$, a contradiction. □

Corollary 5.6. `Tail.OneLine.headSet_one_line_branch`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$ and set $H = H(A, a, b)$. If $\|\mathrm{tail}_H(a-\tau b)\|_2 \le \theta(A)/32$ for some $\tau \in \{+1, -1\}$, then $\varepsilon(a, b) \ge c_{\mathrm{line}}(A, B)$.

Proof. Since $|H| \le d^*(A)$, Proposition 5.5 applies. □

6. The diffuse branch

In the diffuse case, both $\mathrm{tail}_H(a \pm b)$ carry substantial ℓ^2 energy, spread across many coordinates each of size at most $\lambda(A)$. The goal is to show that X_a and X_b are both bounded away from zero with positive probability, which forces $|X_a^2 - X_b^2|$ to be large.

The argument has three layers. First, we partition the tail coordinates into blocks of comparable energy. Second, on the product space $\Omega \times \Omega$, the symmetrized block sums satisfy a small-ball bound via Sperner's antichain inequality. Third, a head–tail factorization converts the anticoncentration into a lower bound on ε.

6.1. Greedy block partition.

Lemma 6.1 (Block partition). `OneD.Blocks.exists_block_partition_with_bounds`

Let $(a_i)_{i=1}^n$ be real numbers satisfying $s > 0$, $s \le \sum_{i=1}^n a_i^2$, and $a_i^2 \le \eta(A, s)$ for all i. Then there exist pairwise disjoint $G_1, \dots, G_{M(A)} \subseteq \{1, \dots, n\}$ with

$$\eta(A, s) \le \sum_{i \in G_j} a_i^2 \le 2\,\eta(A, s) \quad \textit{for } j = 1, \dots, M(A).$$

Proof. Build the blocks greedily. The total energy is at least $s = 4M\eta$. At each step, when m blocks remain, the unused energy is at least $2m\eta$, so a nonempty qualifying subset exists. Choosing a minimal qualifying subset S with $\sum_S a_i^2 \ge \eta$: if this sum exceeds 2η, dropping any one element still leaves at least η (since each $a_i^2 \le \eta$), contradicting minimality. □

6.2. **Anticoncentration on the product space.** Fix a coefficient vector $c \in \mathbb{R}^n$ supported on H^c, and let $G_1, \ldots, G_M$ be the block partition from Lemma 6.1. Pass to the product space $(\Omega \times \Omega, \mu \otimes \mu)$ and define the differences $D_i(\omega_1, \omega_2) = \xi_i(\omega_1) - \xi_i(\omega_2)$, which are independent and symmetric (Theorem A.2). The block sums $Y_j = \sum_{i \in G_j} c_i D_i$ inherit both properties.

Lemma 6.2 (Block mass). `OneD.FiniteConcentration.block_two_sided_mass`

Let $\xi_1, \ldots, \xi_n$ be independent, centered, unit-variance random variables with $\mathbb{E}[|\xi_i|] \geq A > 0$. Let $(c_i)_{i \in G_j}$ satisfy $\eta \leq \sum_{G_j} c_i^2 \leq 2\eta$, and define $Y_j = \sum_{G_j} c_i D_i$ on $\Omega \times \Omega$. Set $a_0 = (A/16)\sqrt{\eta}$. Then

$$(\mu \otimes \mu)\big(|Y_j| \geq a_0\big) \geq A^2/256.$$

Proof. Let $W = \sum_{G_j} c_i \xi_i$. The two-sided mass bound (Theorem A.12) gives $\mathbb{P}(W \geq (A/16)\sqrt{\eta}) \geq A^2/256$ and likewise for $-W$. Since $\mathbb{P}(W \geq 0) + \mathbb{P}(W \leq 0) \geq 1$, combining the events $\{W(\omega_1) \geq (A/16)\sqrt{\eta}\} \times \{W(\omega_2) \leq 0\}$ and its mirror yields the stated bound on $\{|Y_j| \geq a_0\}$. □

Lemma 6.3 (Small-ball bound). `OneD.FiniteConcentration.smallBallMass_sum_symmetric_le`

Let $Y_1, \ldots, Y_M$ be independent symmetric random variables on $\Omega \times \Omega$ with $M = M(A) \geq 256$. Suppose $(\mu \otimes \mu)(|Y_j| \geq a_0) \geq A^2/256$ for each j, and let $0 < r < a_0$. Then for every $t \in \mathbb{R}$,

$$(\mu \otimes \mu)\Big(|\textstyle\sum_{j=1}^M Y_j - t| \leq r\Big) \leq 1/8.$$

Proof. Let $N = \#\{j : |Y_j| \geq a_0\}$ and split:

$$\mathbb{P}\Big(|\textstyle\sum_j Y_j - t| \leq r\Big) \leq \mathbb{P}\Big(|\textstyle\sum_j Y_j - t| \leq r,\ N \geq 256\Big) + \mathbb{P}(N < 256).$$

On $\{N \geq 256\}$, at least 256 of the Y_j exceed $a_0 > r$ in absolute value, so the Sperner sign-flip bound (Theorem A.16) gives a probability at most $1/16$. For the second term: the expected count $\sum_j \mathbb{P}(|Y_j| \geq a_0) \geq M \cdot A^2/256 \geq 1024$, so the Chebyshev bound (Theorem A.10) gives $\mathbb{P}(N < 256) \leq 1/16$. □

Proposition 6.4 (Symmetrized small-ball bound). `OneD.FiniteConcentration.symmetrized_small_ball_le`

Let $\xi_1, \ldots, \xi_n$ be independent, centered, unit-variance random variables with $A \leq \mathbb{E}[|\xi_i|] \leq B$ for $0 < A \leq B < 1$. Let $c \in \mathbb{R}^n$ satisfy $s > 0$, $s \leq \sum c_i^2$, and $|c_i| \leq \lambda_0(A, s)$ for all i. Then for every $t \in \mathbb{R}$,

$$\mu\big(|X_c - t| \leq r_0(A, s)\big) \leq 3/8,$$

where $X_c = \sum_{i=1}^n c_i \xi_i$.

Proof. Unsymmetrization (Theorem A.1) gives

$$\mu\big(|X_c - t| \leq r_0\big)^2 \leq (\mu \otimes \mu)\big(|X_c(\omega_1) - X_c(\omega_2)| \leq 2r_0\big).$$

Write $X_c(\omega_1) - X_c(\omega_2) = \sum_j Y_j + T$, where T is the residual outside the blocks. Since $\sum_j Y_j$ and T act on disjoint coordinates, the concentration function satisfies $\mathrm{Q}(\sum_j Y_j + T, 2r_0) \leq \mathrm{Q}(\sum_j Y_j, 2r_0)$ (`Imported.L2Theory.concFn_add_indep_le`). One checks that $r_0 < a_0$ and $M(A) \geq 256$, so Lemma 6.3 gives $\mathrm{Q}(\sum_j Y_j, 2r_0) \leq 1/8$. Hence $\mu(|X_c - t| \leq r_0) \leq \sqrt{1/8} < 3/8$. □

6.3. **Joint anticoncentration and the cross estimate.** Applying Proposition 6.4 to two coefficient vectors simultaneously, we obtain:

Corollary 6.5 (Cross anticoncentration). `OneD.Cross.cross_anticoncentration_fin`

Under the hypotheses of Proposition 6.4, let $c, d \in \mathbb{R}^n$ both satisfy $s > 0$, $s \leq \sum c_i^2$, $s \leq \sum d_i^2$, and $|c_i|, |d_i| \leq \lambda_0(A, s)$. Then

$$\mathbb{P}\big(r_0(A, s) < \min(|X_c|,\ |X_d|)\big) \geq 1/4.$$

Proof. At $t = 0$, Proposition 6.4 gives $\mathbb{P}(|X_c| \le r_0) \le 3/8$ and $\mathbb{P}(|X_d| \le r_0) \le 3/8$. The union bound gives $3/8 + 3/8 = 3/4$, so the complement has probability at least $1/4$. □

It follows that the minimum of $|X_c|$ and $|X_d|$ has positive second moment, uniformly over all shifts:

Lemma 6.6 (Cross minimum-square bound). `OneD.Cross.cross_anticoncentration_fin`

Let $c, d \in \mathbb{R}^n$ satisfy the hypotheses of Corollary 6.5 with $s = 8\, s_0(A)$. Then for all $u, v \in \mathbb{R}$,

$$\big(r_0(A,\, 8s_0)/4\big)^2 \le \mathbb{E}\big[\min\big(|X_c + u|,\, |X_d + v|\big)^2\big].$$

Proof. Corollary 6.5 applied with shifts $-u, -v$ gives $\mathbb{P}(r_0 < \min(|X_c+u|, |X_d+v|)) \ge 1/4$. Writing $f = \min(|X_c + u|, |X_d + v|)$, this means $\mathbb{E}[f] \ge r_0/4$ (Markov), and $\mathbb{E}[f]^2 \le \mathbb{E}[f^2]$ (Jensen). □

The final ingredient converts anticoncentration of the tail into a lower bound on ε, by factoring $X_a^2 - X_b^2$ into head and tail contributions:

Lemma 6.7 (Head–tail factorization). `OneD.Cross.finEps_ge_two_cross_min_sq`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$ and let $H \subseteq \{1, \dots, n\}$. Define tail-supported vectors

$$c_i = \frac{a_i - b_i}{\sqrt{2}}, \quad d_i = \frac{a_i + b_i}{\sqrt{2}} \quad \text{for } i \notin H, \qquad c_i = d_i = 0 \text{ for } i \in H.$$

If there exists $r > 0$ with $r \le \mathbb{E}[\min(|X_c + u|,\, |X_d + v|)^2]$ for all $u, v \in \mathbb{R}$, then $\varepsilon(a, b) \ge 2r$.

Proof. The identity $X_a^2 - X_b^2 = X_{a+b} \cdot X_{a-b}$ and the head–tail splitting give

$$X_{a+b} \cdot X_{a-b} = 2\,\big(X_d + h_+/\sqrt{2}\big)\big(X_c + h_-/\sqrt{2}\big),$$

where $h_\pm = X_{\mathrm{head}_H(a\pm b)}$ are the head sums. Thus

$$|X_a^2 - X_b^2| \ge 2\,\min\big(|X_c + h_-/\sqrt{2}|,\, |X_d + h_+/\sqrt{2}|\big)^2.$$

Since X_c, X_d are supported on H^c and $h_\pm$ on H, they are independent (`Imported.L2Theory.indepFun_finLinComb_disjoint`). Conditioning on $h_\pm$ and applying the hypothesis gives $\varepsilon(a, b) \ge 2r$. □

Proposition 6.8 (Diffuse cross branch). `OneD.Cross.diffuse_cross_branch_fin`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$ and $H = H(A, a, b)$. If

$$16\, s_0(A) \le \|\mathrm{tail}_H(a - b)\|_2^2 \;\text{ and }\; 16\, s_0(A) \le \|\mathrm{tail}_H(a + b)\|_2^2,$$

and $\max(|(\mathrm{tail}_H\, a)_i|, |(\mathrm{tail}_H\, b)_i|) \le \lambda(A)$ for all i, then $\varepsilon(a, b) \ge c_{\mathrm{cross}}(A)$.

Proof. The rotated coefficients c, d from Lemma 6.7 satisfy $\sum c_i^2 \ge 8s_0$, $\sum d_i^2 \ge 8s_0$, and $c_i^2 \le 2\lambda^2 \le \lambda_0(A, 8s_0)^2$. Lemma 6.6 with $s = 8s_0$ gives $\mathbb{E}[\min^2] \ge (r_0/4)^2$ for all shifts, and Lemma 6.7 converts this to $\varepsilon \ge 2(r_0/4)^2$. Unwinding the definitions:

$$2\big(r_0(A,\, 8s_0)/4\big)^2 = \frac{A^8}{2^{55}} \ge \frac{A^8}{2^{57}} = c_{\mathrm{cross}}(A).$$ □

7. Proof of the main theorem

Theorem 7.1 (Epsilon lower bound). `Main.UniformAB.orthogonal_eps_lower_bound`

Let $a, b \in \mathbb{R}^n$ be unit vectors with $\langle a, b\rangle = 0$, and let $\xi_1, \dots, \xi_n$ be independent random variables with $\mathbb{E}[\xi_i] = 0$, $\mathbb{E}[\xi_i^2] = 1$, and $A \le \mathbb{E}[|\xi_i|] \le B$ for $0 < A \le B < 1$. Then

$$\varepsilon(a, b) \ge \min\big(c_{\mathrm{cross}}(A),\, c_{\mathrm{line}}(A, B)\big).$$

Proof. Set $H = H(A, a, b)$ and apply the dichotomy (Lemma 3.1). In the diffuse case, Proposition 6.8 gives $\varepsilon \ge c_{\mathrm{cross}}$ (the tail coefficient bound follows from the definition of H). In the concentrated case, Corollary 5.6 gives $\varepsilon \ge c_{\mathrm{line}}$. □

Proof of Theorem 1.1. Theorem 7.1 gives $\varepsilon(a,b) \geq \min(c_{\text{cross}}, c_{\text{line}})$, and the bridge inequality (Theorem A.6) gives $\varepsilon(a,b) \leq 2\,\delta(a,b)$. Combining:

$$\delta(a,b) \geq \frac{\varepsilon(a,b)}{2} \geq \frac{1}{2}\min(c_{\text{cross}}, c_{\text{line}}) = c_{\perp}(A,B). \qquad \square$$

Appendix A. Toolbox: Statements Used in the Main Proof

This appendix collects the statements from the "imported" layer that are invoked by the core proof (Sections 5–7). All statements refer to a probability space (Ω, μ) with independent random variables $\xi_1, \ldots, \xi_n$, and use the notation $X_c = \sum_{i=1}^n c_i \xi_i$ for coefficient vectors $c \in \mathbb{R}^n$. Proofs are available in the Lean formalisation.

A.1. **Symmetrization.**

Theorem A.1 (Unsymmetrization). `Imported.Symmetrization.unsymmetrization_product_space`

Let $W : \Omega \to \mathbb{R}$ be a square-integrable random variable on a probability space (Ω, μ). Then for every $r > 0$ and every $t \in \mathbb{R}$,

$$\mu\big(|W - t| \leq r\big)^2 \leq (\mu \otimes \mu)\big(|W(\omega_1) - W(\omega_2)| \leq 2r\big).$$

Theorem A.2 (Differences preserve independence). `Imported.Symmetrization.iIndepFun_differences`

If $\zeta_1, \ldots, \zeta_n$ are independent random variables on (Ω, μ), then the differences $D_i(\omega_1, \omega_2) = \zeta_i(\omega_1) - \zeta_i(\omega_2)$ are independent on $\Omega \times \Omega$.

Theorem A.3 (Desymmetrization L^1 bound). `Imported.Symmetrization.desymmetrization_l1_bound`

For any integrable random variable $W : \Omega \to \mathbb{R}$,

$$\mathbb{E}_{\mu\otimes\mu}\big[|W(\omega_1) - W(\omega_2)|\big] \leq 2\,\mathbb{E}_{\mu}\big[|W|\big].$$

A.2. L^2 **Theory.**

Theorem A.4 (Concentration drops under independent addition). `Imported.L2Theory.concFn_add_indep_le`

Let U and R be independent random variables on (Ω, μ), and let $r > 0$. Then $\mathrm{Q}_\mu(U + R, r) \leq \mathrm{Q}_\mu(U, r)$, where $\mathrm{Q}_\mu(W, r) = \sup_{t\in\mathbb{R}} \mu(|W - t| \leq r)$.

Theorem A.5 (L^2 norm of finite linear combinations). `Imported.L2Theory.randL2Sq_finLinComb_eq`

Let $\xi_1, \ldots, \xi_n$ be independent, centered, unit-variance random variables, and let $c = (c_1, \ldots, c_n) \in \mathbb{R}^n$. Then $\mathbb{E}\big[|X_c|^2\big] = \sum_{i=1}^n c_i^2$.

Theorem A.6 (Epsilon–modulus inequality). `Imported.L2Theory.finEps_le_two_mul_finModulusGap`

Let $a, b \in \mathbb{R}^n$ with $\|a\|_2 = \|b\|_2 = 1$. Then $\varepsilon(a,b) \leq 2\,\delta(a,b)$.

Theorem A.7 (Jensen's inequality). `Imported.L2Theory.sq_integral_le_integral_sq`

For any square-integrable random variable W on (Ω, μ), $\big(\mathbb{E}[|W|]\big)^2 \leq \mathbb{E}[W^2]$.

Theorem A.8 (Independence of disjoint supports). `Imported.L2Theory.indepFun_finLinComb_disjoint`

Let $c, d \in \mathbb{R}^n$ be coefficient vectors with $\mathrm{supp}(c) \cap \mathrm{supp}(d) = \emptyset$, where $\mathrm{supp}(c) = \{i : c_i \neq 0\}$. Then X_c and X_d are independent.

A.3. **Indicator Tails.**

Theorem A.9 (Chebyshev indicator tail). `Imported.IndicatorTails.chebyshev_indicator_tail`

Let $Z_1, \dots, Z_m$ be independent $\{0,1\}$-valued random variables with $\sum_{j=1}^m \mathbb{E}[Z_j] \geq 1024$. Then $\mathbb{P}\big(\sum_{j=1}^m Z_j < 256\big) \leq 1/16$.

Theorem A.10 (Chebyshev active count). `Imported.IndicatorTails.chebyshev_active_count`

Let $Y_1, \dots, Y_m$ be independent random variables, and let $\alpha > 0$. If $\sum_{j=1}^m \mathbb{P}(|Y_j| \geq \alpha) \geq 1024$, then $\mathbb{P}\big(\#\{j : |Y_j| \geq \alpha\} < 256\big) \leq 1/16$.

A.4. **Elementary Bounds.**

Theorem A.11 (Finite independent sum L^1 lower bound). `Imported.ElementaryBounds.finite_indep_sum_l1_lower`

Let $\zeta_1, \dots, \zeta_n$ be independent, mean-zero, unit-variance random variables with $\mathbb{E}[|\zeta_i|] \geq A > 0$ for all i. Then for every $(c_1, \dots, c_n) \in \mathbb{R}^n$,

$$\frac{A}{4}\sqrt{\sum_{i=1}^n c_i^2} \;\leq\; \mathbb{E}\Big[\Big|\sum_{i=1}^n c_i\zeta_i\Big|\Big].$$

Theorem A.12 (Two-sided mass of mean-zero sum). `Imported.ElementaryBounds.two_sided_mass_of_mean_zero`

Let $\zeta_1, \dots, \zeta_n$ be independent, mean-zero, unit-variance random variables with $\mathbb{E}[|\zeta_i|] \geq A > 0$ for all i, and let $(c_1, \dots, c_n) \in \mathbb{R}^n$ with $\sigma = \sqrt{\sum c_i^2} > 0$. Then

$$\mathbb{P}\Big(\tfrac{A}{16}\,\sigma \leq \sum_{i=1}^n c_i\zeta_i\Big) \geq A^2/256.$$

A.5. **Rademacher: Sign Invariance.**

Theorem A.13 (Sign-flip integral equality). `Imported.Rademacher.SignInvariance.sign_flip_integral_eq`

Let $Z_1, \dots, Z_n$ be independent and symmetric random variables, and let $(c_1, \dots, c_n) \in \mathbb{R}^n$. Then for any fixed sign pattern $\varepsilon \in \{+1, -1\}^n$,

$$\mathbb{E}\Big[\Big|\sum_{i=1}^n c_i\varepsilon_i Z_i\Big|\Big] = \mathbb{E}\Big[\Big|\sum_{i=1}^n c_i Z_i\Big|\Big].$$

A.6. **Rademacher: Symmetric Sums.**

Theorem A.14 (Symmetric Khintchine). `Imported.Rademacher.SymmetricSums.symmetric_khintchine_l1_l2`

Let $Z_1, \dots, Z_n$ be independent and symmetric random variables with $\mathbb{E}[Z_i^2] = 1$ and $\mathbb{E}[|Z_i|] \geq A > 0$ for all i. Then for every $(c_1, \dots, c_n) \in \mathbb{R}^n$,

$$\frac{A}{2}\sqrt{\sum_{i=1}^n c_i^2} \;\leq\; \mathbb{E}\Big[\Big|\sum_{i=1}^n c_i Z_i\Big|\Big].$$

A.7. **Rademacher: Sperner.**

Theorem A.15 (Sperner antichain bound). `Imported.Rademacher.Sperner.weighted_sign_interval_antichain`

Let $w_1, \ldots, w_n > 0$ with $w_i \geq \alpha > 0$ for all i, and let I be an interval of diameter less than 2α. Then

$$\#\Big\{\varepsilon \in \{+1,-1\}^n : \sum_{i=1}^{n} \varepsilon_i w_i \in I\Big\} \;\leq\; \binom{n}{\lfloor n/2 \rfloor}.$$

A.8. **Rademacher: Random Sperner.**

Theorem A.16 (Sperner sign-flip). `Imported.Rademacher.RandomSperner.sperner_sign_flip_bound`

Let $Y_1, \ldots, Y_m$ be independent and symmetric random variables, let $\alpha > 0$, $r > 0$ with $r < \alpha$, and let $m \geq 256$. Then for every $t \in \mathbb{R}$,

$$\mu\Big(\Big\{|\textstyle\sum_{j=1}^{m} Y_j - t| \leq r\Big\} \cap \Big\{\#\{j : |Y_j| \geq \alpha\} \geq 256\Big\}\Big) \leq 1/16.$$

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